Health Analytics Dashboard Using MongoDB: A NoSQL Approach to Multi-Source Wearable Data Integration

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***Abstract*—Wearable fitness technologies such as WHOOP and Fitbit continuously generate large volumes of semi-structured physiological, activity, and sleep data. Integrating heterogeneous datasets from multiple devices into a single analytical platform is difficult with traditional relational databases because of rigid schema requirements and evolving data structures. This paper presents a Health Analytics Dashboard that unifies the Fitbit public dataset (35 users, approximately two months) and personal WHOOP data (approximately 2.5 years) in a MongoDB Atlas cluster, connected to an interactive Streamlit visualization layer. A Python-based ETL pipeline cleans, standardizes, and ingests all data into five MongoDB collections. The dashboard delivers activity analysis, sleep performance analytics, physiological insights, and cross-platform comparisons through six interactive tabs. The implementation shows that MongoDB's flexible document model, cloud scalability, and schema-less design are well suited to multi-source IoT health data and offer practical advantages over a relational design for this use case.**

***Index Terms*—MongoDB, NoSQL, Document Databases, Wearable Devices, Internet of Things, Health Informatics, ETL, Streamlit, Data Visualization**

I. Introduction

Wearable fitness technologies such as WHOOP and Fitbit are IoT devices that continuously collect physiological, workout, and health-related data. These devices generate large volumes of stored data that help users better understand their own bodies. The Fitbit public dataset contains daily records from 35 users over approximately two months, while the WHOOP export contains the author's personal data from the past 2.5 years. As a result, the combined data is diverse, semi-structured, and irregular.

Integrating datasets from heterogeneous wearable sources into a single analytical system presents a significant technical challenge: the data do not conform to a uniform structure. Traditional relational database management systems (RDBMS) enforce rigid schemas, fixed table definitions, and relational constraints, which makes them a poor fit for data whose structure continuously evolves. Wearable datasets frequently introduce new metrics, contain missing values, and record observations at inconsistent frequencies.

This paper addresses these challenges by merging Fitbit public data and personal WHOOP data into a unified platform that supports comparative analysis, sleep performance analytics, physiological insights, and interactive visualization. MongoDB [2] is selected as the database backend for its flexible document-oriented schema, horizontal scalability, and cloud-hosted deployment through MongoDB Atlas, which avoid SQL join complexity and schema migration overhead. The primary contributions of this work are:

1) a multi-source wearable data integration architecture using MongoDB Atlas;
2) a Python-based ETL pipeline for cleaning, standardizing, and ingesting heterogeneous CSV data;
3) a six-tab Streamlit dashboard providing interactive health analytics; and
4) an evaluation of the advantages of NoSQL storage for evolving, semi-structured IoT health data.

II. Background and Related Work

A. Wearable Fitness Technologies

Modern wearable health devices capture a broad spectrum of biometric signals. Fitbit devices record daily step counts, calorie expenditure, the distribution of activity intensity, and sleep duration [1]. WHOOP, a higher-resolution wearable, captures Heart Rate Variability (HRV), Resting Heart Rate (RHR), strain scores, recovery percentages, sleep performance, and workout data, including activity type and duration [3].

The heterogeneity of these sources—differing naming conventions, timestamp formats, measurement granularities, and schema structures—makes unified analysis non-trivial. Surveys of IoT data management note the difficulty of forcing such diverse, high-velocity data into fixed relational schemas [4], [5].

B. NoSQL Databases for IoT Health Data

Document-oriented NoSQL databases such as MongoDB are increasingly recognized as appropriate storage backends for IoT sensor data because of their schema flexibility, horizontal scalability, and native support for JSON-like documents [2], [6]. Unlike relational systems, MongoDB permits documents within the same collection to contain different fields, enabling

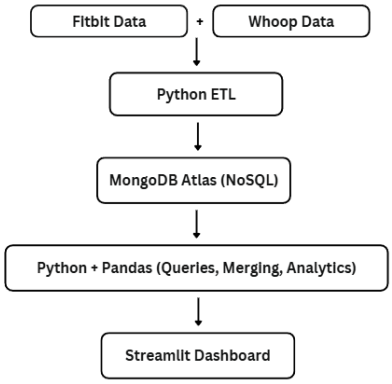


Fig. 1. System architecture and data flow.

TABLE I
DATA SOURCES

Source / Coverage	Files
Fitbit (public) 35 users, ∼2 months	`daily_activity.csv` `sleep_day.csv`

WHOOP (personal)
1 user, ∼2.5 years
	`Physiological_rafi.csv`
	`Sleeps_rafi.csv`
	`workouts_rafi.csv`

multiple wearable data schemas to coexist without table alterations or null-padded columns.

MongoDB Atlas extends self-hosted MongoDB with cloud-managed deployment, automated scaling, built-in monitoring, and high availability through replica sets [2]. Prior work has examined NoSQL stores for scalable IoT data management [4], [5], motivating their adoption in the present wearable health context.

III. System Architecture and Design

The proposed system ingests, stores, and analyzes multi-source wearable data through a four-component pipeline: data sources, ETL processing, MongoDB Atlas storage, and a Streamlit visualization layer. Fig. 1 illustrates the end-to-end data flow.

A. Data Sources

Two primary data sources are incorporated, as summarized in Table I. The Fitbit public dataset, sourced from Kaggle [1], includes the `daily_activity.csv` and `sleep_day.csv` files, covering 35 anonymized users over approximately two months in 2016. The WHOOP personal dataset was exported directly from the WHOOP mobile application and comprises three files spanning approximately 2.5 years of continuous wear.

B. ETL Processing Pipeline

Raw CSV files from both sources are cleaned, transformed, and standardized using Python and the Pandas library. The ETL process involves: (i) converting ISO timestamp strings to normalized date objects; (ii) computing aggregate metrics such as daily averages and sleep efficiency ratios; (iii) removing irrelevant or redundant columns; (iv) imputing or flagging null values; and (v) deriving custom metrics, including strain-normalized recovery scores and 6-month period aggregates. Sleep efficiency is computed as

$$\text{Sleep Efficiency} = \frac{T_{\text{asleep}}}{T_{\text{in bed}}} \times 100\% \tag{1}$$

where T_{asleep} is total minutes asleep and $T_{\text{in bed}}$ is total minutes in bed. Each processed record is inserted into the appropriate MongoDB collection as a JSON document.

C. MongoDB Atlas Database Layer

Processed data is stored in a MongoDB Atlas cluster deployed as a three-node replica set on AWS (us-east-1), as shown in Fig. 2. Five collections are maintained: `daily_activity`, `sleep_day`, `physiologicals_rafi`, `sleeps_rafi`, and `workouts_rafi`. MongoDB's flexible document model accommodates the divergent schemas of Fitbit and WHOOP data within the same cluster without enforcing a rigid unified schema. Atlas provides real-time monitoring of read/write operations, connection counts, and storage utilization, enabling performance oversight throughout dashboard operation.

D. Visualization Layer

The Streamlit dashboard retrieves data from MongoDB via PyMongo queries and performs in-memory transformations—including date-based merging, monthly and 6-month interval grouping, and metric computation—before rendering interactive Altair charts. The dashboard is deployed on Streamlit Cloud with GitHub integration for continuous delivery.

IV. Implementation

A. ETL Workflow and Schema Alignment

The principal ETL challenge involved aligning two datasets that record similar health constructs—such as sleep duration and caloric expenditure—using incompatible field names, timestamp granularities, and measurement semantics. Timestamps in WHOOP exports contain time-zone offsets and sub-second precision; these were stripped to date-only keys to enable join operations with Fitbit's date-formatted records. Mismatched date ranges were handled by filtering to overlapping periods when cross-device comparisons were required.

Schema alignment was achieved without enforcing a unified document structure in MongoDB; instead, application-layer merging in Pandas is performed at query time in the Streamlit dashboard. This preserves MongoDB's schema flexibility while enabling ad hoc comparative analysis.

B. MongoDB Collection Design

Each wearable data stream is stored in a dedicated collection, preserving source-specific document structures (Table II). Documents in `daily_activity` contain Fitbit fields, while those in `physiologicals_rafi` contain WHOOP fields. Because there is no rigid unified schema, new metrics introduced by firmware updates can be added to future documents

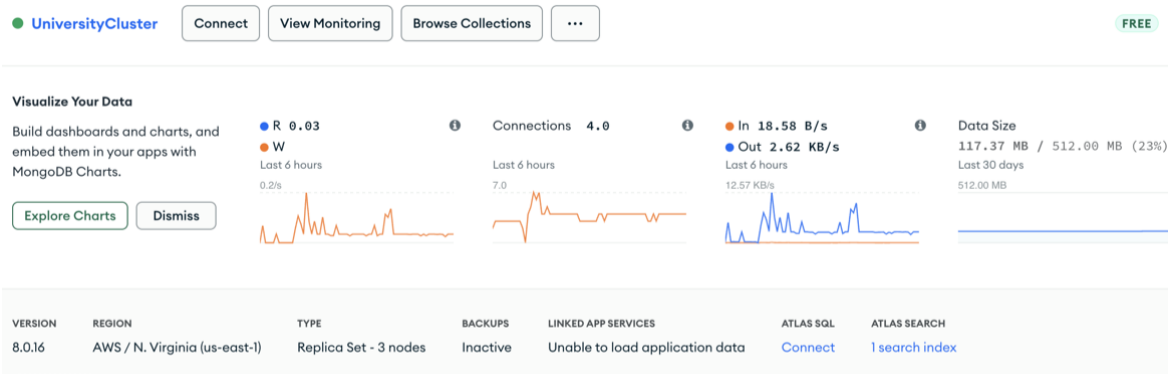


Fig. 2. MongoDB Atlas cluster overview showing operations, connections, network traffic, and data size.

TABLE II
MONGODB COLLECTIONS

Collection	Source	Example Fields
`daily_activity`	Fitbit	TotalSteps, Calories, VeryActiveMinutes
`sleep_day`	Fitbit	TotalMinutesAsleep, TotalTimeInBed
`physiologicals_rafi`	WHOOP	HRV, RHR, SkinTemp, BloodOxygen
`sleeps_rafi`	WHOOP	Sleep need, performance
`workouts_rafi`	WHOOP	Activity, duration, strain

without migrating existing records—a key advantage over relational storage for longitudinal IoT datasets.

C. Dashboard Structure and Analytics

The Streamlit application is organized into six tabs: Project Overview, Fitbit Activity, Fitbit Sleep Activity, WHOOP Activity, WHOOP Sleep Activity, and Comparisons (Fig. 3). Each tab retrieves the relevant MongoDB collection, applies the necessary aggregation, and renders Altair visualizations.

The Fitbit tabs include a multi-user selector covering all 35 users, giving each individual personalized activity and sleep views. The Fitbit Activity tab visualizes steps, calories, intensity composition, and user summaries. The Fitbit Sleep tab displays sleep duration trends, weekday versus weekend differences, sleep efficiency, and daily patterns. The WHOOP Activity tab displays HRV and RHR trends, monthly physiological averages, strain–energy relationships, and workout intensity. The WHOOP Sleep tab provides sleep performance segments and monthly sleep trends. The WHOOP tabs expose longitudinal trends unavailable within the native WHOOP application, which limits visualization to the most recent six months. A dedicated Comparisons tab highlights cross-platform differences in calories, sleep quality, and duration.

V. SETUP METHODOLOGY

The dashboard was assessed on three criteria: (i) its ability to unify Fitbit and WHOOP datasets in a single store; (ii) whether it produces meaningful activity, sleep, and physiological analysis; and (iii) whether the implementation demonstrates the practical advantages of MongoDB for semi-structured health data. The software stack consisted of Python, Pandas, PyMongo, Altair, Streamlit, and MongoDB Atlas.

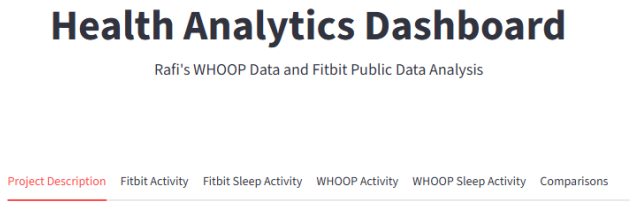


Fig. 3. Dashboard header and six-tab navigation.

VI. RESULTS AND FINDINGS

A. Fitbit Activity Analysis

Analysis of the Fitbit dataset shows a clear positive relationship between total daily step count and caloric expenditure (Fig. 4). Activity intensity composition, visualized per user as a pie chart (Fig. 5), shows that most daily activity time is spent in low-intensity or sedentary states. For the representative user 1503960366, high- and moderate-intensity activity combined accounts for less than 15% of recorded daily minutes, in line with population-level evidence on the prevalence of physical inactivity [7].

B. Fitbit Sleep Analysis

Average sleep duration is consistently longer on weekend days than on weekdays among the examined Fitbit users (Fig. 6), a pattern consistent with research on social jetlag and weekend sleep recovery [8]. No formal significance test was performed, so this is reported as a descriptive trend. Day-to-day sleep duration, visualized as a time series (Fig. 7), exhibits periodic spikes corresponding to nights of extended recovery.

C. WHOOP Physiological and Activity Analysis

Analysis of 2.5 years of personal WHOOP data shows HRV fluctuating with noticeable monthly peaks, indicating periodic recovery (Fig. 8). Average RHR remains stable



Fig. 4. Steps versus calories burned (Fitbit, single user).

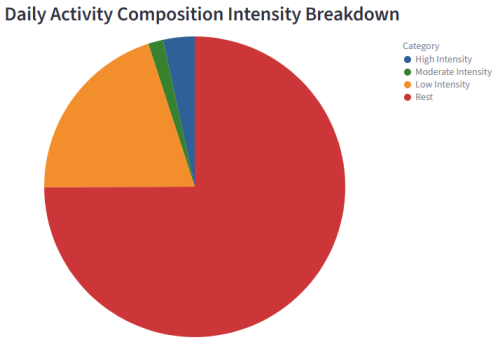


Fig. 5. Daily activity composition by intensity (user 1503960366).

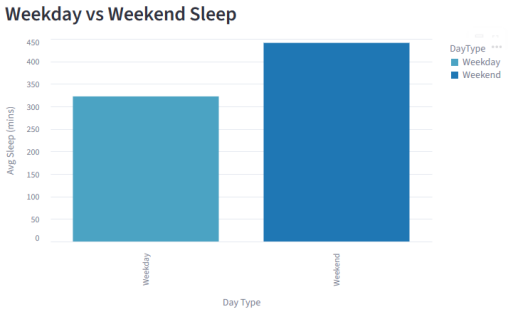


Fig. 6. Average sleep duration, weekday versus weekend (Fitbit).

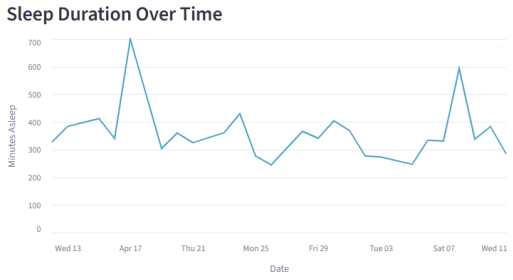


Fig. 7. Sleep duration over time (Fitbit).

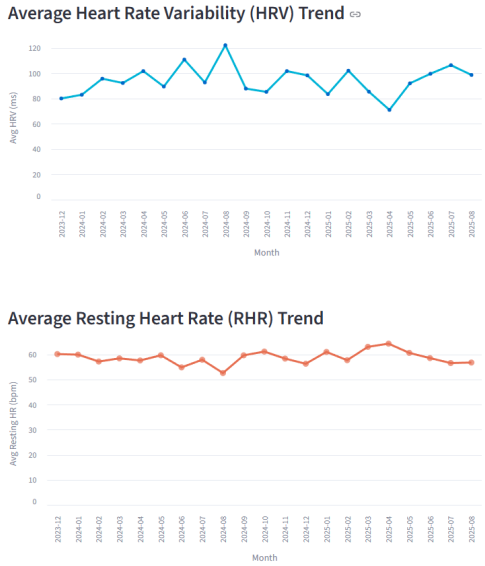


Fig. 8. Monthly average HRV (top) and RHR (bottom) trends (WHOOP).

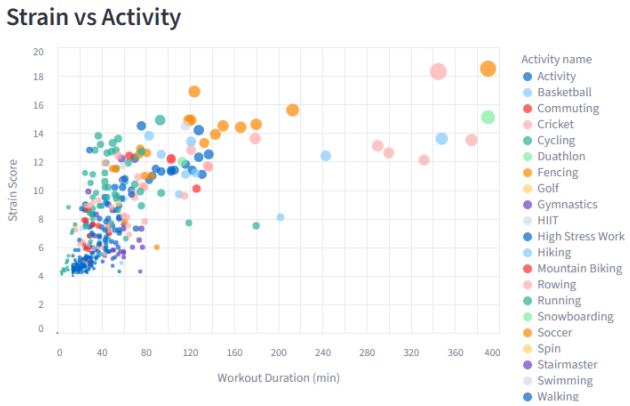


Fig. 9. Strain score versus workout duration by activity type (WHOOP).

across the observation period (mean approximately 53 bpm), suggesting consistent cardiovascular conditioning. The strain versus workout duration chart (Fig. 9) shows that endurance activities—including running, cycling, and cardio sessions—generate substantially higher strain scores than recreational activities, consistent with the expected dose–response relationship between exercise intensity and physiological load.

D. WHOOP Sleep Analysis

WHOOP sleep summary metrics (Table III, Fig. 10) indicate an average sleep need of 9.3 hours with an average sleep performance of 61%, reflecting habitual sleep insufficiency relative to physiological requirement. Semi-annual aggregates show a peak in sleep duration in the second half of 2024 and the first half of 2025, followed by a decline in mid-2025. Sleep performance follows a similar trajectory (Fig. 11). These multi-year trends are not visible in the native WHOOP

TABLE III
WHOOP SLEEP SUMMARY METRICS

Metric	Value
Avg. sleep needed	9.3 h
Avg. sleep performance	61%
Best night	100%
Worst night	7%

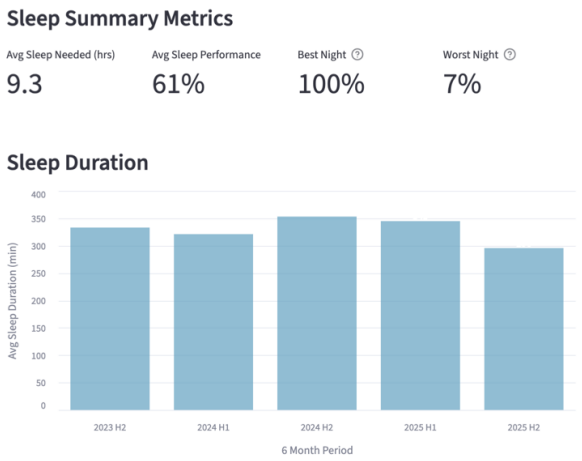


Fig. 10. Sleep summary metrics and average sleep duration per 6-month period (WHOOP).

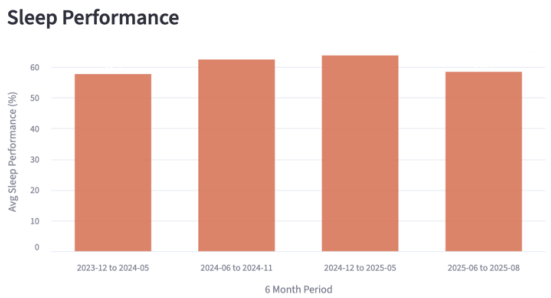


Fig. 11. Average sleep performance per 6-month period (WHOOP).

application, which truncates historical visualization to six months, underscoring the value of the custom MongoDB-backed dashboard. All sleep metrics are computed directly from documents in the WHOOP collections, where each entry is stored as its own document.

E. Cross-Platform Comparison

The Comparisons tab reveals notable differences between Fitbit and WHOOP metrics. The WHOOP user exhibits a marginally higher average daily caloric expenditure than the Fitbit cohort mean, while Fitbit users record longer average sleep durations. Sleep efficiency, as measured by each device's algorithm, is higher in the Fitbit dataset than in the WHOOP dataset. These comparisons should be read with care: they contrast a single WHOOP user with a 35-user cohort, recorded years apart, using proprietary and non-equivalent device algorithms. They are nonetheless enabled directly by MongoDB's document-oriented architecture: Fitbit and WHOOP records, stored as independent documents in separate collections, are merged at the application layer without schema unification at the database level.

VII. DISCUSSION

A. Suitability of MongoDB for Wearable Health Data

The results confirm that MongoDB's flexible document model is well suited to multi-source wearable health data. Three properties are particularly advantageous. First, schema flexibility allows Fitbit and WHOOP documents to coexist in the same cluster with different field sets, avoiding the null padding and schema migration overhead a relational design would require. Second, timestamped document storage naturally accommodates the append-only write pattern of IoT sensor data, where new daily records are continuously inserted without modifying historical documents. Third, MongoDB Atlas's cloud-hosted architecture provides high availability through replica set replication and offloads infrastructure management, enabling focus on application-layer analytics.

B. Challenges Encountered

Several technical challenges arose during development. Data cleaning required resolving incompatible timestamp formats, field naming conventions, and measurement semantics between Fitbit and WHOOP exports. Integrating MongoDB Atlas with Streamlit Cloud required secure credential management, PyMongo connection configuration, and iterative debugging of network compatibility. Visualization complexity arose from reshaping MongoDB query results into Altair-compatible dataframes, particularly for multi-period aggregations. Deployment stability required correct GitHub integration and dependency pinning to ensure reproducible Streamlit Cloud builds.

C. Limitations

This study did not benchmark MongoDB against a relational database, so its advantages are argued from design and implementation experience rather than measured performance. The WHOOP data represents a single user, and the Fitbit data dates from 2016, which limits the generality of cross-platform findings.

VIII. FUTURE WORK

Several directions for enhancement are identified. First, integrating the Fitbit Web API [1] and the WHOOP API [3] would eliminate reliance on manual CSV exports and enable continuous automated ingestion. Second, migrating analytic computations from Pandas into MongoDB aggregation pipelines would leverage server-side processing for improved scalability and more fully demonstrate advanced NoSQL capabilities. Third, machine learning models trained on the longitudinal WHOOP dataset could estimate optimal recovery scores and sleep requirements. Fourth, incorporating additional platforms, such as Apple Health and Garmin Connect, would broaden cross-device comparison. Fifth, a quantitative benchmark against a

relational baseline would strengthen the comparative claims. Finally, user authentication and multi-tenancy would allow the dashboard to serve multiple users, each viewing only their own data.

IX. Conclusion

This paper presented a Health Analytics Dashboard that integrates Fitbit public data and personal WHOOP data into a unified MongoDB Atlas-backed platform, demonstrating the practical advantages of NoSQL document storage for multi-source IoT health data. The system's ETL pipeline, flexible schema design, and Streamlit visualization layer together enable comparative sleep analysis, longitudinal physiological trend detection, and cross-platform metric comparison that would be more complex to implement with a relational database. The WHOOP longitudinal analysis, in particular, surfaces multi-year trends unavailable in the native application.

The project reinforced key NoSQL principles—schema flexibility, document-oriented storage, cloud scalability, and ETL-driven data integration—through a practical, data-intensive application. Future extensions targeting real-time API ingestion, server-side aggregation pipelines, and predictive modeling are expected to further demonstrate MongoDB's capabilities for health informatics workloads.

The deployed application and source code are available at:
https://mdrafi4632-health-analytics-dashboard-.streamlit.app
https://github.com/Mdrafi4632/health_analytics_dashboard

Acknowledgment

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